Imbalance Mitigation for TCGA-UT Across Patch and WSI-Bag Regimes

Abstract

Class imbalance in computational pathology is not a single evaluation problem: labeled image patches support ordinary image classification, whereas whole-slide image (WSI) classification is commonly formulated as weakly supervised multiple-instance learning over patch-feature bags. This study evaluates imbalance mitigation on TCGA-UT with two native-regime benchmarks rather than a single cross-regime leaderboard. The patch benchmark compares cross-entropy, class weighting, focal loss, balanced sampling, Scholz-style combined losses (CE with soft F1 or soft MCC plus balanced sampling), and ProGAN-based synthetic augmentation. The WSI-bag benchmark compares matched attention-MIL baselines with RankMix and SC-MIL. All methods use slide-disjoint splits, repeated three-seed evaluation, and shared backbones within each benchmark. On patches, plain cross-entropy gives the highest macro F1 (0.314 ± 0.004), while balanced sampling gives the highest balanced accuracy (0.333 ± 0.013) at lower macro F1. On WSI bags, RankMix is the strongest method, improving balanced accuracy over plain MIL CE from 0.836 ± 0.020 to 0.863 ± 0.008 and macro F1 from 0.840 ± 0.020 to 0.848 ± 0.017. The results show that advanced imbalance mechanisms are not uniformly beneficial, but can help when their assumptions match the native input regime.

1 Introduction

Class imbalance changes what a pathology classifier is rewarded for learning. When common tumor types dominate training data, high aggregate accuracy can coexist with unreliable rare-class recall, especially if evaluation emphasizes prevalence-sensitive summaries. Long-tailed learning and medical-image evaluation literature therefore motivate classwise recall, macro F1, balanced accuracy, and calibration-aware summaries as primary outcomes in imbalanced settings. [1–3]

A common source of ambiguity in pathology imbalance studies is the input regime. Some methods are naturally image-level methods: they assume labeled images, image embeddings, or directly generated image samples. Other methods are naturally WSI-level methods: they assume weak labels over bags of instances and use attention, instance ranking, or bag-level representation learning. Comparing such methods in one pooled leaderboard can confound the imbalance mechanism with changes in supervision, input size, backbone, and aggregation.

TCGA-UT is useful for separating these questions because it supports two legitimate tasks with different native inputs. The dataset provides labeled histopathology patches cropped from pathologist-selected tumor regions and slide identities that can be grouped into WSI-level bags. Patch labels are class labels, not dense pixel masks or cell-level annotations. This study therefore uses the available supervision directly: image-level methods are evaluated on controlled patch classification, while MIL methods are evaluated on slide-level feature bags.

The contribution is a controlled benchmark design rather than a reproduction claim for every cited method. The study uses one dataset, two native tasks, and one controlled comparison within each task. Method fidelity is defined by implementing the defining imbalance mechanism in the regime for which that mechanism was designed, while holding the backbone family, split, and evaluation code fixed within each benchmark.

2 Dataset and Shared Splits

The dataset contains 8736 slides across 32 cancer-type labels. Class support is long-tailed: the largest class contains 792 slides, the smallest contains 28 slides, giving a slide-level imbalance ratio of 28.3. For each seed, slides are split into 6116 training, 1310 validation, and 1310 test slides. The same slide-level assignments are reused by both benchmarks so that no patches from the same slide can appear in different splits.

The patch benchmark derives a deterministic controlled patch manifest from the shared slide split. It uses resolution level 0 and samples 10 patches per slide with a fixed rule before method-specific training begins. This makes the patch benchmark a controlled comparison of imbalance mechanisms rather than a comparison of patch-selection policies. The WSI-bag benchmark uses frozen Virchow2 feature bags built from the same slide identities. Profiling on one GPU found bags with 10–70 instances, a mean of 30.8 instances, and a profiled batch of 32 bags requiring about 55 MB of CUDA peak memory for one forward/backward pass. The configured WSI experiments therefore used full bags rather than an arbitrary instance cap.

3 Methods

The study organizes methods into four imbalance-mitigation categories: data-level augmentation, synthetic generation, algorithm-level losses, and representation or architecture methods. The central control is comparability within regime. The patch benchmark uses one shared patch-classification backbone, and the WSI-bag benchmark uses one shared attention-MIL backbone. A method is named after a cited paper only where its defining imbalance mechanism is implemented in the corresponding native regime.

3.1 Patch-Level Benchmark

Table 1: Patch-level benchmark methods.

Category	Method	Implemented mechanism
Baselines	CE, weighted CE, focal loss, balanced sampling	Shared patch classifier with objective or sampler changes only.
Algorithm-level	CE + soft F1/MCC (balanced)	Equal-weight sum of cross-entropy with macro soft F1 or multiclass soft MCC, trained with class-balanced oversampling. [4]
Synthetic generation	ProGAN augmentation	Class-specific progressive generators balance minority patch classes to the training-set head count; this is a Ruiz-Casado-style ProGAN adaptation from binary to multiclass histology. [5]

The patch baselines isolate common imbalance interventions. Weighted CE changes the class contribution to the loss, focal loss down-weights easy examples, and balanced sampling changes minibatch composition without changing the loss. The Scholz representatives follow the authors' strongest configuration: class-balanced oversampling together with an equal-weight combination of cross-entropy and a differentiable macro soft-F1 or multiclass soft-MCC objective. ProGAN augmentation trains class-specific generators for minority classes and augments the patch training set; generated patches are used only as additional patch-classification training examples, not as synthetic slide bags.

Table 2: WSI-bag benchmark methods.

Category	Method	Implemented mechanism
Baselines	MIL CE, weighted MIL, focal MIL, balanced MIL	Shared attention-MIL backbone with objective or sampler changes only.
Data-level augmentation	RankMix	Two-stage attention-MIL training with teacher-derived patch pseudo-labels, ranked representative feature subsequences, and soft-label feature mixup. [6]
Representation	SC-MIL	Bag-level supervised contrastive learning with a projector head and a linear curriculum from representation to classifier learning. [7]

3.2 WSI-Bag Benchmark

The WSI baselines use the same attention-MIL architecture and frozen feature bags. RankMix first trains a teacher MIL model, derives ranked instance subsets from the teacher attention and pseudo-label signal, and then trains with mixed representative feature subsequences and soft labels. SC-MIL adds a supervised-contrastive bag representation objective through a projection head and linearly shifts training weight toward the classifier objective over time. Activation diagnostics are recorded for both methods to verify that augmentation and contrastive pair construction occur during training.

4 Evaluation

Both benchmarks use seeds 0, 1, and 2. Macro F1 is the primary ranking metric within each benchmark because it weights classes equally and penalizes failure on rare classes. Balanced accuracy, accuracy, classwise precision/recall/F1, head/body/tail summaries, negative log likelihood, multiclass Brier score, and expected calibration error are also computed where probabilities are available. Patch and WSI-bag methods are never ranked against each other, because they answer related but different questions with different input units and supervision.

All reported tables use the held-out test split and show mean $\pm$ standard deviation over seeds. Separate validation summaries are retained as artifacts for model-selection checks. The completed sweep contains no missing planned results for either benchmark.

5 Results

5.1 Patch-Level Results

Table 3: Patch-level test results over three seeds. Methods share the same patch manifest, resolution, classifier family, and evaluation code.

Method	Accuracy	Balanced accuracy	Macro F1
CE	0.412 ± 0.011	0.310 ± 0.003	0.314 ± 0.004
Focal	0.402 ± 0.010	0.307 ± 0.011	0.310 ± 0.011
ProGAN augmentation	0.407 ± 0.012	0.306 ± 0.021	0.299 ± 0.021
Balanced sampler	0.354 ± 0.011	0.333 ± 0.013	0.293 ± 0.006
Weighted CE	0.333 ± 0.033	0.330 ± 0.016	0.285 ± 0.018

Plain CE is the strongest patch method by the primary metric, with macro F1 0.314 ± 0.004 and accuracy 0.412 ± 0.011 (Table 3). Focal loss is close, with macro F1 0.310 ± 0.011. Balanced sampling and weighted CE shift the operating point toward rare-class sensitivity: they produce the highest balanced accuracies, 0.333 ± 0.013 and 0.330 ± 0.016, but lower macro F1 and accuracy. This tradeoff indicates that the class-balancing interventions improved average recall but introduced enough precision or head-class loss to reduce the harmonic classwise summary.

Among the representative advanced patch methods, ProGAN augmentation reaches macro F1 0.299 ± 0.021, below CE and focal loss (Table 3). The Scholz combined losses are reported in the same table; their macro-F1 and balanced-accuracy rankings should be read against both plain CE and balanced-sampling CE, because the Scholz configuration deliberately couples metric-derived objectives with oversampling. The result should not be interpreted as a general rejection of synthetic generation or metric-derived losses. It is narrower: under this fixed patch set, classifier family, and three-seed configuration, not every imbalance mechanism that helps in other medical imaging settings transfers to TCGA-UT patches.

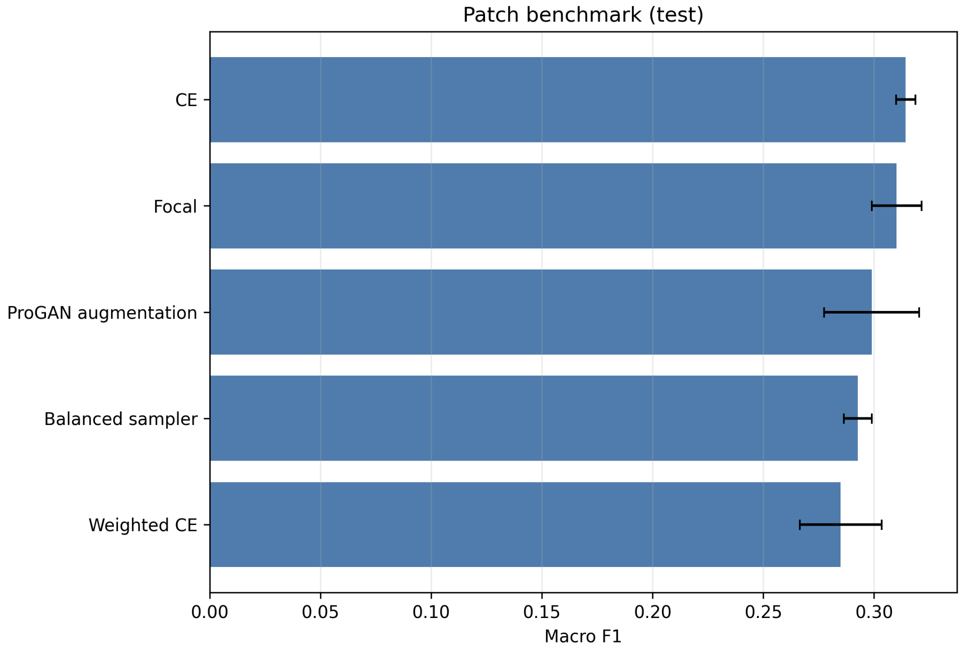

Figure 1: Patch-level macro F1 on the held-out test split. The figure visualizes the same ranking as Table 3; absolute values should only be interpreted within the patch benchmark.

5.2 WSI-Bag Results

Table 4: WSI-bag test results over three seeds. Methods share the same full feature bags, attention-MIL backbone family, and evaluation code.

Method	Accuracy	Balanced accuracy	Macro F1
RankMix	0.894 ± 0.010	0.863 ± 0.008	0.848 ± 0.017
MIL CE	0.890 ± 0.006	0.836 ± 0.020	0.840 ± 0.020
Balanced-sampling MIL	0.885 ± 0.007	0.837 ± 0.015	0.836 ± 0.013
SC-MIL	0.881 ± 0.007	0.837 ± 0.017	0.833 ± 0.016
Weighted MIL CE	0.880 ± 0.005	0.834 ± 0.018	0.829 ± 0.018
Focal MIL	0.879 ± 0.004	0.824 ± 0.011	0.827 ± 0.009

RankMix is the strongest WSI-bag method by all three aggregate metrics (Table 4). Relative to plain MIL CE, it improves balanced accuracy from 0.836 ± 0.020 to 0.863 ± 0.008 and macro F1

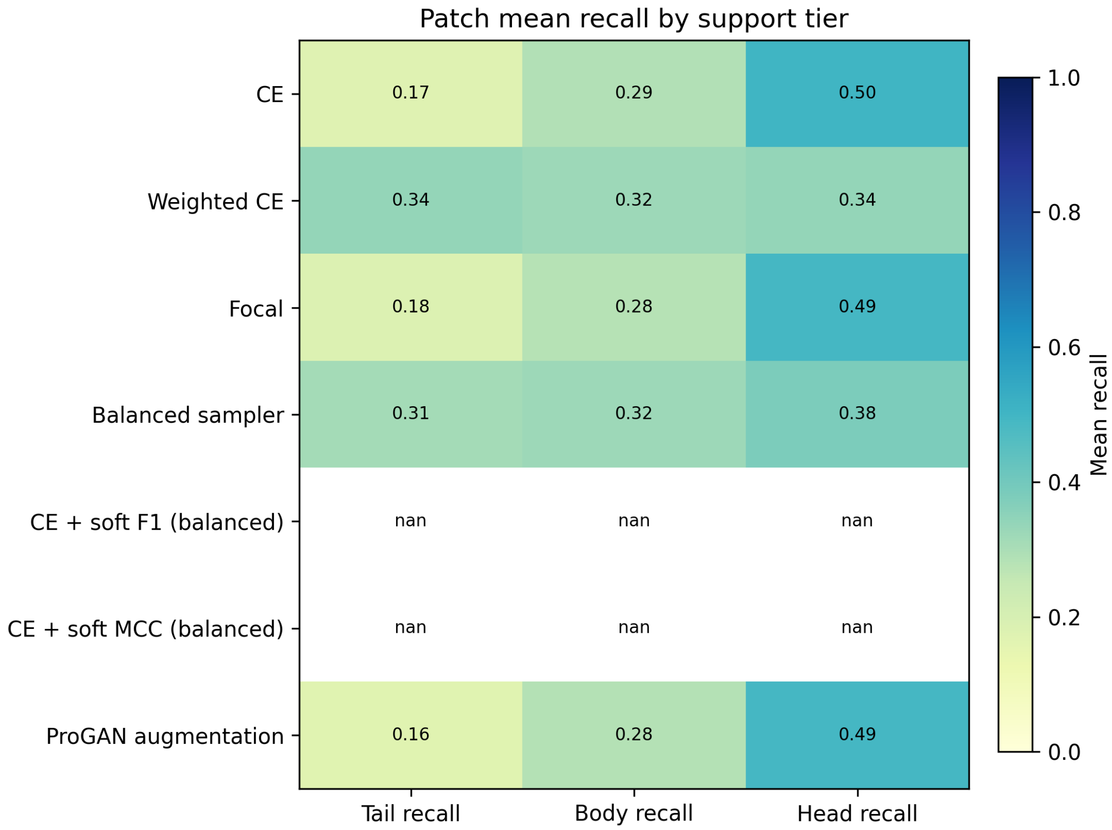

Figure 2: Patch-level classwise recall on the held-out test split. The plot exposes the class-level variation hidden by aggregate accuracy and motivates reporting balanced accuracy and macro F1.

from 0.840±0.020 to 0.848±0.017. The macro-F1 gain is modest, but the balanced-accuracy gain is larger and consistent with RankMix directly augmenting representative instance subsequences in the MIL regime.

The remaining WSI methods cluster closely. Balanced-sampling MIL and SC-MIL produce similar balanced accuracy (0.837), but both trail RankMix in macro F1. SC-MIL performs well on validation but does not exceed plain MIL CE on the test macro-F1 average. Weighted MIL CE and focal MIL do not improve over plain MIL CE in this configuration, with focal MIL giving the lowest WSI macro F1. Activation diagnostics confirm that the advanced WSI objectives were active: RankMix generated about 1.10 million mixed examples per seed, and SC-MIL constructed about 1.06 million positive pairs per seed.

6 Discussion

The two benchmarks support different conclusions. In the patch benchmark, simple CE and focal loss are hard to beat by macro F1, while class-balanced sampling and weighting mainly trade macro F1 for balanced accuracy; metric-derived Scholz losses are evaluated in that same tradeoff space because they are trained with balanced sampling. In the WSI-bag benchmark, RankMix improves the strongest baseline and is the clearest case where a more elaborate imbalance-specific mechanism helps. The same study therefore mixes negative and positive transfer results across patch-level synthetic or metric-derived methods and WSI-native feature mixing.

This pattern reinforces the need to evaluate imbalance methods in their native input regimes. A patch generator should not be judged by first converting synthetic images into one-instance

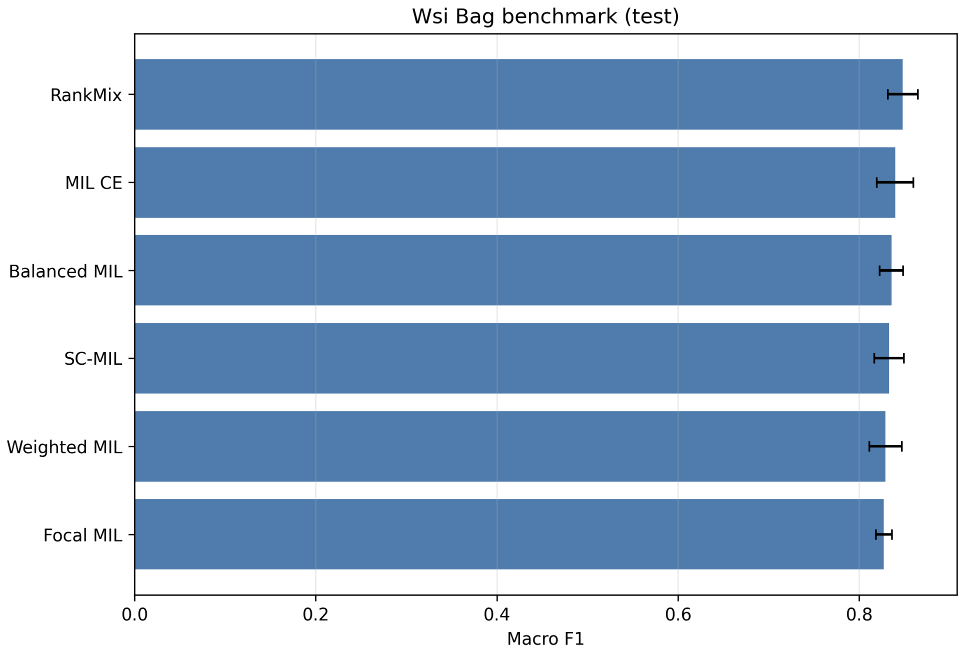

Figure 3: WSI-bag macro F1 on the held-out test split. RankMix is the top method within the WSI-bag benchmark, but the absolute scale is not comparable to the patch benchmark because the task input differs.

MIL bags, and a MIL feature-mixing method should not be compared directly against a patch classifier. The controlled design narrows the claims but makes them easier to defend: within each benchmark, the split, backbone family, input budget, and evaluation code are shared.

Several limitations remain. First, the study is a controlled benchmark, not a full reproduction of every original paper. The ProGAN experiment keeps the progressive-generator mechanism from Ruiz-Casado-style histology augmentation but adapts it from binary breast histology to multiclass TCGA-UT and does not reproduce the full GAN-family comparison. Second, TCGA-UT patch labels come from pathologist-selected tumor regions; they are useful patch-level class labels but not dense annotations. Third, the WSI benchmark uses frozen Virchow2 feature bags and an attention-MIL family, so conclusions apply to this representation setting. Finally, advanced methods may benefit from additional hyperparameter tuning beyond the fixed three-seed benchmark used here.

7 Conclusion

TCGA-UT supports a cleaner imbalance study when patch-level and WSI-bag supervision are evaluated separately. On controlled patch classification, the strongest macro-F1 baseline is plain CE, and balancing interventions mainly change the recall–precision tradeoff. On WSI bags, RankMix provides the strongest overall result and the clearest improvement over a plain MIL baseline. The practical implication is that imbalance mitigation should be selected and evaluated at the same level where the model actually consumes data; otherwise, apparent method differences can be artifacts of incompatible task regimes.

References

[1] M. Saini and S. Susan, "Tackling class imbalance in computer vision: a contemporary review," *Artificial Intelligence Review*, vol. 56, no. S1, pp. 1279–1335, 2023.

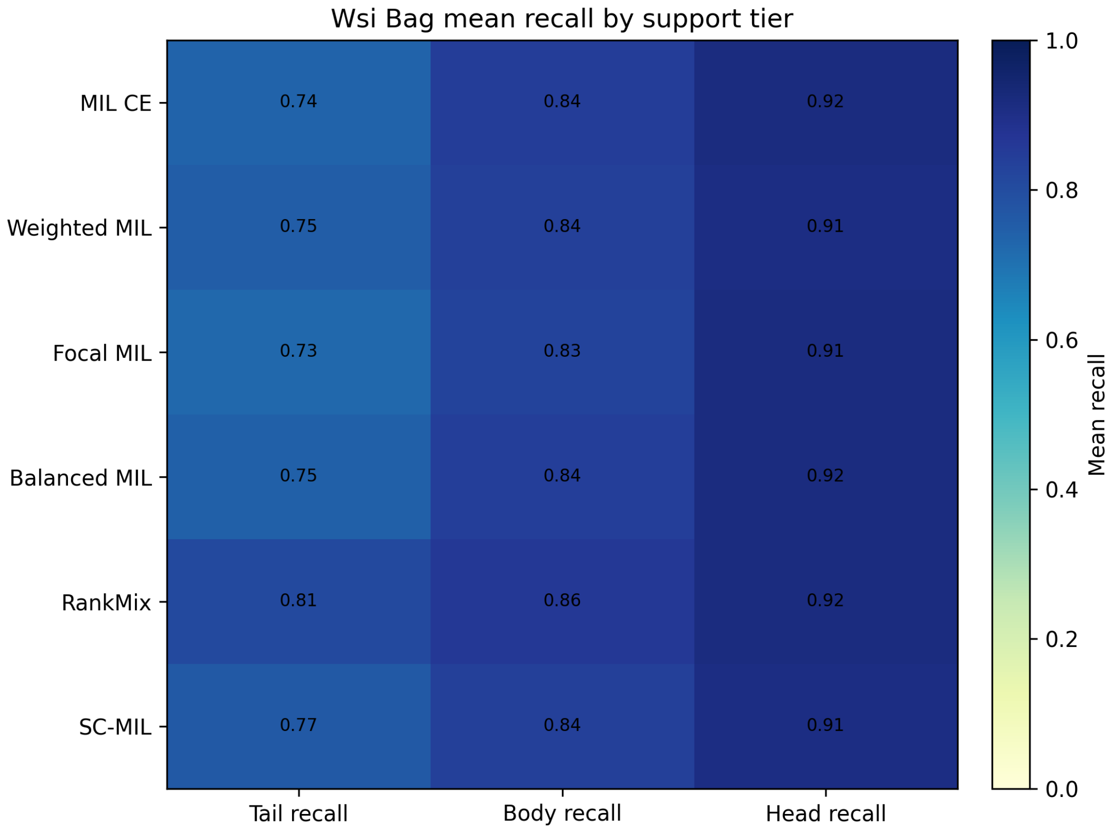

Figure 4: WSI-bag classwise recall on the held-out test split. The spread across classes shows why aggregate accuracy alone is insufficient for the long-tailed setting.

[2] Y. Zhang, B. Kang, B. Hooi, and S. Yan, “Deep long-tailed learning: A survey,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 9, pp. 10795–10816, 2023.

[3] C. Mosquera, L. Ferrer, D. H. Milone, D. Luna, and E. Ferrante, “Class imbalance on medical image classification: towards better evaluation practices for discrimination and calibration performance,” *European Radiology*, vol. 34, no. 12, pp. 7895–7903, 2024.

[4] D. Scholz, A. C. Erdur, J. Buchner, J. C. Peeken, D. Rueckert, and B. Wiestler, “Imbalance-aware loss functions improve medical image classification,” in *Proceedings of the 7th International Conference on Medical Imaging with Deep Learning*, vol. 250 of *Proceedings of Machine Learning Research*, pp. 1341–1356, PMLR, 2024.

[5] J. L. Ruiz-Casado, M. A. Molina-Cabello, and M. R. Luque-Baena, “Enhancing histopathological image classification performance through synthetic data generation with generative adversarial networks,” *Sensors*, vol. 24, no. 12, p. 3777, 2024.

[6] Y.-C. Chen and C.-S. Lu, “Rankmix: Data augmentation for weakly supervised learning of classifying whole slide images with diverse sizes and imbalanced categories,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 23936–23945, 2023.

[7] D. Juyal, S. Shingi, S. A. Javed, H. Padigela, C. Shah, A. Sampat, A. Khosla, J. Abel, and A. Taylor-Weiner, “Sc-mil: Supervised contrastive multiple instance learning for imbal-

anced classification in pathology," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2024.